RIFQI HABIB UR RAHMAN

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SUMMARY

Experienced Embedded Systems Engineer with hands on experience, Skilled in Embedded Systems Hardware and software. Also certified in Cisco CCNA1 and KKNI Level 2 Teknik Komputer Jaringan. Proficient in English and Bahasa Indonesia.

WORK EXPERIENCE

Operational Intern, Ryu Logium **Oct 2021 - Dec 2021**

- Keeping track of IDR to USD exchange rate.
- Operate accounting software.
- Taking care of customs documents and requirements.
- Manage both physical and digital document archives

Part-time Robotics Lecturer, Syafana Primary School **Sept 2024 - Present**

- Taught primary school students on Robotics and block Coding (in english).
- Structure a syllabus for the lectures.

Embedded Systems Engineer, Smart Trap Pilot Project, MySalak (EPICS in IEEE) **Aug 2025 - Present**

- Developed embedded software for automated pest-monitoring devices.
- Designed and assembled the hardware using STM32 and Seeed XIAO ESP32-C3 microcontrollers.
- Implemented long-range communication between traps and gateway using LoRa and LoRaWAN, ensuring reliable data transmission in field conditions.
- Coordinated end-to-end prototype development, Lab Testing, field testing, and technical documentation.

RELEVANT ACTIVITIES

Sustainability Innovation Challenge 2024 (SIC2024) - Participant **Sept 2024**

- Worked with students from Nanyang Polytechnics (Singapore) and Multimedia University (Malaysia) in a group project setting.
- Went through a mentoring session with several Company representatives from Singapore, Malaysia, as well as Indonesia.

Australian Youth Innovation Challenge 2025 - Semi-Finalist **June 2025**

- Pitched SensoRice (Rice paddy IoT infrastructure).
- Became a speaker at Diversity Innovation Tech Festival 2025 (DITF 2025) at Perth City Hall, WA.
- Worked on the hardware of SensoRice, Specifically the water level adjustment node.

PROJECTS

Arduino UNO Midi Controller **Mar 2024**

- Developed a midi controller compatible with most DAW
- reflashed the arduino uno’s Atmega 16U2 with existing open source USB MIDI Firmware.

SketchMidi Controller **Dec 2024**

- Compatible with most operating system DAW.
- Web based preset controls.
- Based on ESP32S3 + arduino pro micro (for USB MIDI).
- 6 Rotary + 6 Fader (multiplexed) input.
- Web connection via MQTT.
- Preset stored at flash.
- Developed the all the microcontroller firmwares for the project.

Melanoma Image Classification (Prociding, CONMEDIA 2025) **Sept 2025**

- Based on MobileNetV2.
- Designed the activation layer architecture for the project.

MySalak Smart Trap **Aug 2025 - Present**

- Pre-Production Model
  - Prototype developed on Xiao ESP32C3.
  - LoRa for communication to gateway.
  - 2x2 IR Breaker for fly detection.
  - Solar Panel to ensure positive power budget.
  - CN371 Solar Charger Controller.
- Production Model
  - Based on the STMicro STM32U5 microcontroller.
  - Custom PCB
  - LoRaWAN for communication to gateway.
  - Developed a proprietary 8x8 IR Breaker Array for pest detection.
  - Limited production models are being tested in Turi,Sleman.
- Developed both Software, Hardware, and designed the physical case for the project.

EDUCATION

Computer Engineering Undergraduate **Aug 2022 - Present**

Universitas Multimedia Nusantara

- 8th Semester Student

ADDITIONAL INFORMATION

- **Technical Skills:** Microcontroller (Hardware & Software), Network Management, Machine Learning, System Administration, CAD, Web Development (React, Laravel), CAD(FreeCAD), EDA(EasyEDA).
- **Languages:** English(IELTS Band 7.5), Bahasa(Native).
- **Programming Language :** C, Cpp, Python.
- **Certifications:** Cisco CCNAV1, KKNI Level 2 Teknik Komputer Jaringan.
- **Awards/Activities:**
  - Head of R&D Division at ACES Gen14 (Association of Computer Engineering Students).
  - Member of R&D Division at ACES Gen15.
  - Sustainability Innovation Challenge 2024 Participant
  - Australian Youth Innovation Challenge 2025 Semi Finalist
  - UMN’s Symbol of Appreciation 2025 Awardee